



Genetic effects of landscape change on three sympatric turtle species in southern Ontario, Canada

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Abstract

Habitat loss restricts movement between habitats, reducing genetic diversity and increasing wildlife's vulnerability to environmental change. Landscape genetics can help identify how habitat loss and heterogeneity influence genetic diversity and genetic differentiation. For long-lived species such as turtles, long generation times can obscure declines in genetic diversity. Thus, it is important to consider how historic, or “ghost” landscapes influence genetic diversity. This study compared the impact of ghost and current habitat amount, configuration, and latitude on the genetic diversity of the Snapping Turtle (*Chelydra serpentina*), the Blanding's Turtle (*Emydoidea blandingii*) and the Spotted Turtle (*Clemmys guttata*). We leverage genetic data from previous studies that have investigated population structure and provide a comparative synthesis on how landscape change is affecting these sympatric species. Allelic richness is lowest at sites with greater wetland loss exclusively in Snapping Turtles. While latitude showed no effect on genetic diversity across species, isolation by distance was significant. Drivers of genetic differentiation varied by species: Snapping Turtle was influenced by forest cover, Spotted Turtle by historic wetland loss, and Blanding's Turtle by geographic distance. Isolation among species due to landscape change and reduced genetic diversity in Snapping Turtles suggest vulnerability to continued habitat loss. This synthesis using data from previous studies provides novel inference and conservation value due to the previously unexamined impact of landscape heterogeneity and latitude on sympatric species nearing their northern range limit. These findings highlight varied responses to habitat loss and the need for habitat restoration that addresses multiple species' conservation needs.

Introduction

Environmental change driven by anthropogenic activities has triggered a wave of population declines and extinctions comparable in both rate and magnitude to the five previous mass extinctions of Earth's history (Barnosky et al. 2012; Dirzo et al. 2014). Human-caused landscape change has led to the loss and isolation of natural habitats, impacting

the abundance, distribution, and connectivity of wildlife populations (Fahrig 2003). These changes are reflected in a population's genetic structure, often leading to declines in genetic diversity which increase a population's vulnerability and extinction risk (Schlaepfer et al. 2018). Retention of intact habitat, successful dispersal, and effective population connectivity is therefore increasingly important for the persistence of wild populations (Saura et al. 2014). In the face of continued landscape change, understanding how dispersal occurs in modified landscapes is essential to developing effective conservation plans focused on maintaining connectivity, population viability and genetic diversity (Hoban et al. 2021; Keller et al. 2015).

At the population and species level, genetic diversity has been declining at an average rate of about 1% per decade since the mid-19th century and tends to be lowest in areas where human influence is greatest (IPBES 2019; Leigh et al. 2019). The conservation of genetic diversity requires active management, as imperiled populations are not expected to recover on their own (Hoban et al. 2023). Indeed, in

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southern Ontario, Canada, much of the historic forest and wetland habitat has been lost. Extensive deforestation for timber and agriculture began in the mid-1700s and peaked in the mid-19th century when most of the historic forests were cleared (Riley 1994). For turtle species, habitat loss has entailed the conversion of wetlands to agricultural and urban settlements, and an estimated 72% of historic wetlands in the region have been drained or altered (Bardecki 1982; Riley et al. 1994; Snell 1987). Loss of wetland habitat continues, with more recent (1978–2019) wetland habitat loss associated with agricultural tile drainage and road building (Penfound and Vaz 2022). Declines in terrestrial habitat, such as forest cover, continues to decline in southern Ontario (Ontario Biodiversity Council 2024). The legacy of this landscape change may contribute to current patterns of genetic diversity and remains largely unexamined for long-lived species.

The field of landscape genetics explores the influence of landscape composition and configuration on spatial patterns of genetic diversity (Manel et al. 2003; Storfer et al. 2007). Though contemporary landscape features influence gene flow, the legacies of historical landscape features also shape current spatial patterns of genetic diversity (Anderson et al. 2010). Time lags between a change or disturbance in the landscape and its reflection in spatial patterns of genetic diversity can vary (Epps and Keyghobadi 2015; Landguth et al. 2010). Long generation times, overlapping generations (Lloyd et al. 2013), and low dispersal (Nei et al. 1977) are some factors expected to increase the time lag response between a change in landscape composition and configuration, and a detectable genetic response (Blair et al. 2012; Landguth et al. 2010). These demographic traits, in addition to delayed sexual maturity and low juvenile survival in turtles, exacerbate their risk of decline (Congdon et al. 1993; Germano and Bishop 2009). Failures to consider the legacies of past landscapes, or “ghost landscapes” (Anderson et al. 2010; Landguth et al. 2010) can risk misattributing current landscape features as true drivers of spatial patterns of genetic diversity, when in fact historical legacies have persisting effects. It is therefore important to disentangle historical landscape features from current, particularly in long-lived species like turtles.

In addition to the effects of ghost landscapes, other processes such as latitudinal gradients and species traits such as generation time and vagility can influence spatial patterns of genetic diversity. The effects of current landscape structure on genetic diversity can therefore be difficult to distinguish from historical events (Epps and Keyghobadi 2015; Wade and McCauley 1988). In turtle species, a negative relationship is expected between genetic diversity and latitude in post-glacial landscapes due to founder effects during northward range expansion (Galbraith 2008). The relatively low

dispersal abilities, small home ranges, and thermoregulatory constraints exhibited by turtles may limit habitat availability at the northern periphery of their range (Lesbarrères et al. 2014). These life history traits and habitat preferences can also result in differences in spatial patterns of genetic diversity among different species of turtle, and sometimes run counter to expectations from population genetic models (McGaugh 2012; Reid et al. 2017). For example, small and isolated populations of turtle species sometimes exhibit high levels of genetic diversity despite recent demographic bottlenecks (Davy and Murphy 2014; Kuo and Janzen 2004).

In this study, we investigated the relative importance of historic and contemporary landscapes in shaping patterns of genetic diversity in three turtle species. The Snapping Turtle (*Chelydra serpentina*) is listed as a species of Special Concern under the federal Species At Risk Act (SARA; S.C. 2002, c. 29). Snapping Turtles are habitat generalists and are found in a variety of aquatic habitats throughout their range including lakes, ponds, rivers, and wetlands and have also been observed making extensive overland movements (Brown et al. 1994; Obbard and Brooks 1980; Paterson et al. 2012). In contrast, the Blanding’s Turtle (*Emydoidea blandingii*) and the Spotted Turtle (*Clemmys guttata*) are listed as Threatened and Endangered respectively under SARA and are both habitat specialists (Rasmussen and Litzgus 2010; Reid et al. 2017). Blanding’s Turtle exhibit habitat preference for home ranges that include wetlands, slow-flowing rivers and creeks, and artificial channels (Beaudry et al. 2009; Millar and Blouin-Demers 2012; Rowe and Moll 1991) but are known to make extensive use of terrestrial habitat, migrating long distances for mating sites (Edge et al. 2010). Forested areas are often a predictor of habitat use for Blanding’s Turtles, which serve as corridors for movement between wetlands (Attum et al. 2008; Millar and Blouin-Demers 2011; Mui et al. 2015). Spotted Turtle home ranges also include wetlands, with preference for marsh meadows and open and woody wetlands and avoidance of open and exposed uplands (Angoh et al. 2021; Rasmussen and Litzgus 2010). In southern Ontario, these three turtle species exhibit different patterns in spatial genetic structure, with genetic clustering analyses suggesting different locations for a genetic split (Davy 2013). The landscape drivers of these varying patterns of diversity, however, remain unclear.

To better understand the drivers of genetic diversity in these three species, we test a series of hypotheses. We first consider node-based hypotheses, which model the spatial distribution of alleles at the sampling location (nodes) (Wagner and Fortin 2013). These node-based hypotheses test the influence of wetland habitat extent at our sample sites on genetic diversity, comparing how historical and current amounts of wetland habitat explain current patterns of

genetic diversity. We expect populations inhabiting areas with historically larger wetland habitats to exhibit higher levels of genetic diversity, as these wetlands may have previously supported larger populations. Historic (c. 1800) wetland extent may therefore be associated with higher genetic diversity in our specialist species (Spotted and Blanding's), and not in our generalist species. We also test the role of latitudinal gradients, as nodes located further north at the edge of these species' ranges may have lower genetic diversity.

We then consider two link-based hypotheses, which model the probability of gene flow between two patches (the link) and relate to landscape heterogeneity (Wagner and Fortin 2013). We consider both individual-based and deme-based measures of isolation-by-distance. We expect patterns of individual-based isolation-by-distance (IBD; Wright 1943), to be weakest in the most abundant, and relatively vagile Snapping Turtle and strongest in the most sedentary species, the Spotted Turtle.

We investigate the role of historic losses of wetland habitat, geographic distance, and current landscape features, on genetic differentiation. Population genetic structure has been characterized in previous studies in all three species. For Snapping Turtle, at least two genetically distinct populations and four subpopulations were identified (Davy 2013). We hypothesized that long generation times and relatively higher abundance in the Snapping Turtle would prevent detectable genetic differentiation attributable to historic wetland connectivity, geographic distance, and in particular current landscape features. For Blanding's Turtle, previous work has indicated a minimum of four genetic clusters across the study area and reduced genetic diversity in disjunct eastern populations (Davy et al. 2014). Given their relatively higher vagility and proclivity to move between wetland habitats, we hypothesized that decreases in historic wetland habitat would have the greatest influence on genetic differentiation in the Blanding's Turtle. Analyses of population genetic structure in the Spotted Turtle identified at least 6 genetically isolated populations, within which 10 subpopulations are nested (Davy and Murphy 2014). In the Spotted Turtle, the significant genetic structure identified could not be explained by geographic proximity of sites or isolation by distance alone (Davy and Murphy 2014). Relative to the Blanding's Turtle, the Spotted Turtle is less vagile. We therefore hypothesized that geographic distance would influence genetic differentiation, in addition to decreases in wetland habitat connectivity.

Multi-species approaches are increasingly used in conservation planning to avoid over-generalizing landscape effects on animal movement (Richardson et al. 2016). The differing ecological requirements (Gibbons 2003), and demographic characteristics (e.g. long generation time) among species make this approach particularly suitable

for studying freshwater turtles (Anthonysamy et al. 2018; Reid et al. 2017). Although population structure has been described in previous studies for each of the species we examine here, investigating similarities and differences in genetic responses to landscape change among these sympatric species is essential to inform how ongoing landscape change may affect the occurrence and persistence of these species.

Methods

Study area and species

Three sympatric turtle species were sampled throughout southern Ontario (Fig. 1). The Snapping Turtle (*Chelydra serpentina*) is widespread across much of southern Ontario, while the Blanding's Turtle (*Emydoidea blandingii*) is widespread, but its distribution is not continuous across the region (Ontario Nature 2020). For the Spotted Turtle (*Clemmys guttata*), sample sites were selected to represent most of the known extant populations of the species in Canada as of 2008 (COSEWIC 2014). Given the initial sample scheme, we organized our analysis and sites into 10 × 10 km UTM squares to effectively compare species over a common spatial extent. Each sample site for the analysis is one 10 × 10 km grid cell in which at least two turtles were sampled. The size of the sample site represents the upper expected limit for home-range size for all three species (Paterson et al. 2021). In our effort to effectively compare species across a common landscape, some individuals fell outside the 10 km square sample site and were assigned to the nearest sample site that fell within the bounds of the ecoregion. This assured the inclusion of as many genetic samples as possible while also assuring environmental conditions were comparable. Spotted Turtles, the least vagile of the three species, were clustered within the sample sites. Some Snapping and Blanding's Turtles, which typically have larger home ranges, were sampled far outside the grid. The range of re-assigned individuals was initially between 9 and 50 km. We examined outlier individuals based on geographic location and identified those with unusually high or low allelic richness and heterozygosity using box plots. Individuals falling outside the interquartile range were considered outliers and removed from further analysis. Re-sampling the genetic data to test this sampling scheme led to the removal of four individual Snapping Turtles. All analysis on the Snapping Turtle was with this filtered dataset ($n = 163$).

Gradients in elevation and landcover types can influence population connectivity in turtles (Reid et al. 2017). Southern Ontario's Niagara Escarpment is a 735 km long limestone outcrop that is densely forested (Fig. 1). South of

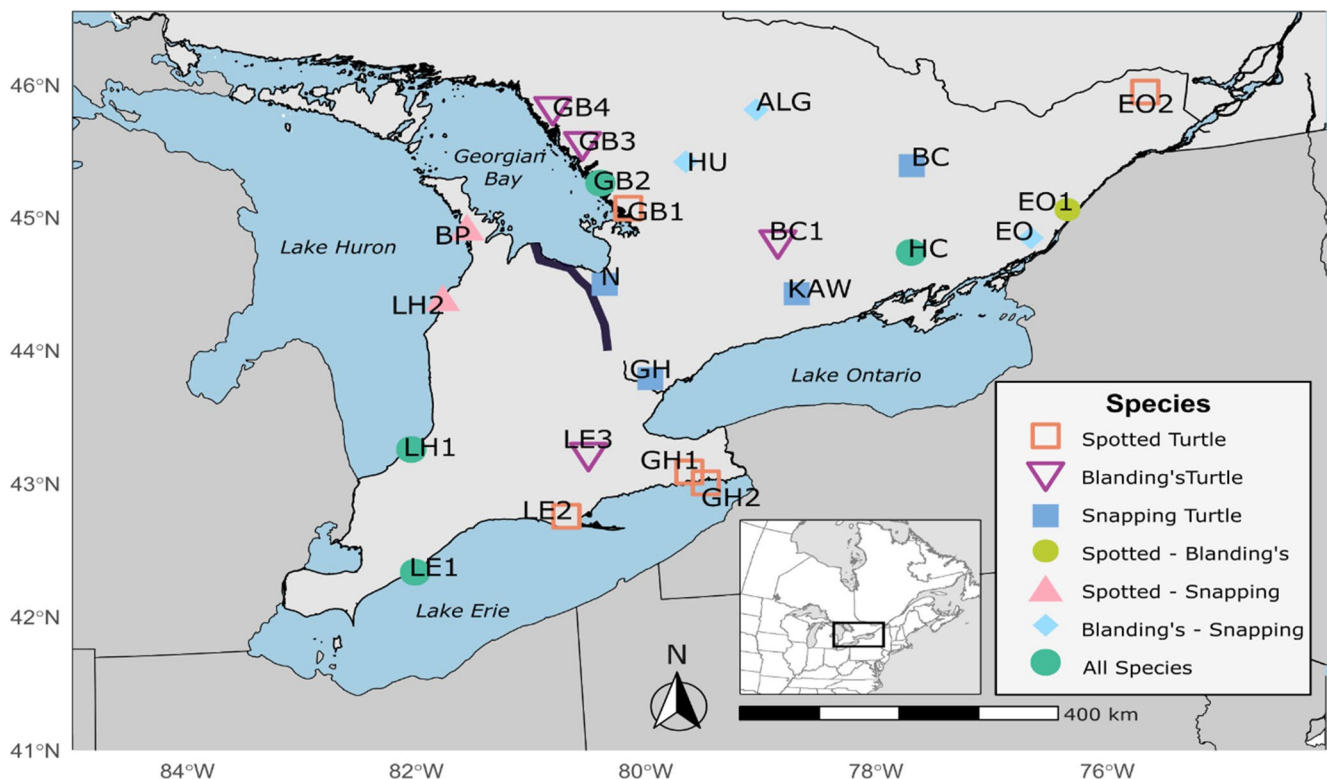


Fig. 1 Sample map of the study area shows sampling of three freshwater turtle species. Niagara Escarpment highlighted with a black line

the Niagara Escarpment the topography is mostly flat and heavily industrialized; much of the region was wetland habitat before conversion to agricultural and settled landscapes. North of the Niagara Escarpment is more forested, and a higher proportion of wetland and forested habitat has been retained. This difference in landcover types is considered in link-based models.

Genetic data

We analyzed three previously published microsatellite datasets of turtle species sampled across southern Ontario, Canada between 2008–2011 (Fig. 1). Turtles were captured either by hand, or in 3-foot hoop traps with 1.5" mesh (TN315 Turtle Nets from Memphis Net and Twine Company, Memphis, U.S.A.) baited with sardines. Blood was sampled by caudal venipuncture with a sterile syringe, and blotted onto FTA cards (Whatman Inc., Clifton, NJ) for storage. Extraction of turtle DNA from FTA cards followed Smith and Burgoyne (2004). At some sample sites, muscle tissue was collected from road-killed individuals and stored in ethanol prior to analysis. For injured individuals rehabilitated at the Kawartha Turtle Trauma Centre (Peterborough, Ontario), blood samples were stored in heparin before analysis. Extraction of DNA from muscle tissue and heparinised

blood followed the phenol-chloroform procedure of Sambrook et al. (1989).

Snapping Turtles ($n=167$) were genotyped at 10 microsatellite loci, based on previously developed loci (Davy et al. 2012). Samples for the Snapping Turtle were amplified at nine loci developed for the Snapping Turtle (Cs08, Cs12, Cs16, Cs17, Cs18, Cs19, Cs22, Cs24, Cs25) and one locus (MteD9) cross-amplified from the Alligator Snapping Turtle (*Macrochelys temminckii*). Further details on the Snapping Turtle dataset can be found in Davy et al. (2012).

Blanding's Turtles ($n=91$) were genotyped at 12 microsatellite loci. Two loci were specifically developed for Blanding's Turtle (*Emydoidea blandingii*)—Eb17 and Eb19 (Osentoski et al. 2002), and an additional 10 loci were amplified from *Glyptemys muhlenbergii* that cross-amplified in Blanding's Turtle (GmuB08, GmuD16, GmuD21, GmuD28, GmuD55, GmuD87, GmuD88, GmuD93, GmuD107 and GmuD121; King and Julian 2004). Further details on the Blanding's Turtle dataset can be found in Davy et al. (2014). Spotted Turtles ($n=256$) were genotyped at 11 loci following the protocol in Davy et al. (2012). Samples for the Spotted Turtle were amplified at eleven microsatellite loci (GmuA19, GmuB08, GmuD16, GmuD21, GmuD55, GmuD79, GmuD87, GmuD88, GmuD107, GmuD114, GmuD121) originally developed for *Glyptemys muhlenbergii* following amplification methods of Schuelke

(2000). Further details on this dataset can be found in Davy and Murphy (2014). DNA extraction for all three species followed Davy et al. (2012). All datasets were tested for deviations from Hardy-Weinberg equilibrium, linkage disequilibrium and null alleles in the previously documented studies. Given the longevity of these species, the genetic samples we used were appropriate for the comparison with available historical wetland data, the “current” of which is circa 2002.

Landcover data

We re-classified the Ontario Landcover Compilation Dataset (Ontario Ministry of Natural Resources and Forestry OLCC v. 2.0, 2016) to test the role of current landcover types: wetland, forest, water, shoreline, agricultural land, and developed land on genetic distance. Variance inflation factors between explanatory landcover variables were calculated in the R package “usdm” (Naimi 2015). To avoid multicollinearity, we fixed a correlation threshold of Pearson $r=0.7$ (Legendre and Legendre 2012), and landscape features in the final models included forest cover, wetland amount, and urban development. Wetland amount and forest cover were not included in the same models due to multicollinearity.

Historic and current wetland data

Historic and current wetland habitat (c. 1800, 1967, 1982, 2002) was derived from Duck’s Unlimited Canada’s (DUC) Southern Ontario Wetland Conversion Analysis (2010) and was augmented with small wetland habitats from the OLCC. The Southern Ontario Wetland Conversion Analysis dataset is a predictive landcover product developed using Landsat imagery, aerial photo interpretation, and spatial modelling to estimate historic wetland extent across southern Ontario. The dataset excludes all natural features smaller than 10 hectares (=0.1% of a 10×10 km grid cell). Swamps (wetland areas dominated by trees or shrubs) and marshes (areas with emergent plants) are largely excluded, as it is difficult to differentiate swamps from lowland forests in the interpretation of aerial data. Swamps are key habitat for Snapping and Blanding’s Turtles, and marshes are important for the Spotted Turtle. To overcome this challenge, we augmented the DUC wetland data with wetland data extracted from the OLCC (c. 2016). This approach assumes that much of the swamp and marsh habitat has persisted and was not recently created. The original DUC dataset is vector data that was rasterized at a resolution of 100 m. The OLCC dataset resolution was 30 m and the *resample* function in the “terra” package in R was used to change the resolution to 100 m.

For node-based analyses, a 10×10 km UTM grid was overlaid and wetland percentage values were extracted using the “landscapemetrics” package v 1.5.4. in R (Hesselbarth et al. 2019).

Influence of wetland habitat amount on genetic diversity at the node-level

In sedentary, specialist species like the Spotted Turtle, dispersal may not be the primary driver of genetic diversity and the amount of wetland habitat at the sample site may better predict genetic variability. Genetic diversity may therefore be best predicted by the amount of habitat at the sampling location (the node). In contrast, Blanding’s and Snapping Turtles are known to disperse greater distances between habitats and an isolation-by-distance or resistance model may be better supported. Genetic diversity in these species may therefore be better predicted by habitat conditions between sites (the link). We used linear regression to model habitat amount (percentage of wetland habitat within the sample site) at different points in time (e.g., 1800, 1967, 1982, and 2002) on genetic diversity for each species. For each species at each sampled site, we calculated expected and observed heterozygosity (H_E , H_O) using the R package “adegenet” (Jombart 2008) and rarefied allelic richness (A_R) using the function *allel.rich* implemented in the R package “PopGenReport” (Adamack and Gruber 2014). Univariate models were run for all time periods (c. 1800, 1967, 1982, 2002) for each species.

Influence of latitudinal gradients on genetic diversity at the node-level

We tested a hypothesis that genetic diversity and connectivity decrease with increasing latitude (Eckert et al. 2008). Genetic diversity was measured using rarefied allelic richness (A_R), expected heterozygosity (H_E), and observed heterozygosity (H_O). We used linear regression to examine the relationship between genetic diversity and latitude of each sample site, using the *lm* function in R (v. 4.0.2). Model assumptions of normal distribution of residuals and homogeneity of variances were visually checked. Residuals vs. leverage plots identified one site exerting influence, which was removed for the Spotted Turtle (GH2, $n=7$). Sites GH2 and GH1 are close together at almost the same latitude, however, site GH1 has a larger sample size (GH2, $n=26$). A_R is much lower at site GH2, and this was driven by sample size. Sites with only two Blanding’s Turtles were also excluded from the analysis, leaving seven remaining sites. The same was done for the Snapping Turtle with one site (HU, $n=2$), as only two individuals were sampled at this site.

and a residuals vs. leverage plot identified it as an outlier exerting influence.

Isolation-by-distance at the link-level

Geographically limited dispersal can result in a correlation between genetic and geographic distance. We calculated the proportion of shared alleles between individuals using the package “PopGenReport” (Adamack and Gruber 2014). Proportion of shared alleles measures the degree of allelic similarity among individuals (Bowcock et al. 1994). At the landscape scale, fine-scale patterns of individual-based IBD are more likely in more dispersal-limited species. Geographic distances were calculated as Euclidean distances between sample locations and were log-transformed (Rousset 1997). Correlations between these distance matrices were calculated using the Mantel test and were implemented in R (v. 4.0.2) using the *mantel* function in the “vegan” package (Dixon 2003) with 999 permutations (Legendre 2012).

Influence of landscape on genetic distance at the link-level

We modeled the relationship between landscape features and genetic distance using maximum likelihood population models (Van Strien et al. 2012) and an information theoretic approach to model selection (Burnham and Anderson 2002). MLPE is a linear regression technique for distance data that controls for the non-independence of pairwise data by using a residual covariance structure within a mixed modeling approach (Clarke et al. 2002). We used a matrix of pairwise measures of genetic distances (F_{ST} ; Weir and Cockerham 1984) calculated in the “hierfstat” package in R for use as the response variable in our model. We first wanted to model the relationship between genetic distance and the resistance based on the amount of wetland habitat on the landscape. For the mostly aquatic generalist Snapping Turtle, we did not expect the loss of wetland habitat to impact genetic structure. For the specialist Blanding’s and Spotted Turtles, we expected that wetland habitat loss and fragmentation would be potentially related to current patterns of spatial genetic structure.

Wetland resistance layers were created using a UTM grid of 1 km resolution over the wetland spatial layers, and the *aggregate* function in the “landscapemetric” package v. 1.5.4. (Hesselbarth et al. 2019) was used to calculate the percentage of wetland within each 1 km grid. Using these data, we calculated multiple resistance surfaces from which least-cost paths (Spear et al. 2010) were derived. A higher percentage of wetland within a grid corresponds to lower

resistance (Cushman and Landguth 2010). The spatial location of least-cost routes is sensitive to differences in relative cost values and is also shaped by the spatial structure of the landscape (Rayfield et al. 2010; Spear et al. 2010). We had no a priori reason to assume any particular relative resistance values for the different land cover types and therefore sought to identify relative values that best captured our hypothesized drivers of turtle movement (Peterman et al. 2014). We first plotted least-cost-paths based on no resistance, to develop isolation-by-distance predictor variables. We then compared these to least-cost-paths based on a conductance surface for the extensive declines in wetland habitat that occurred between 1800 and 2002. We found that least-cost paths in areas of areas of diffuse wetland habitat would sometimes travel through cells without any wetland habitat. To better capture the important differences in wetland cover in these different time periods, we squared the raw numbers of wetland per 1 km cell. The transformation allowed us to better compare the relative effects of wetland cover in 1800 and 2002 and to test our specific hypotheses. Least-cost effective distances between sample locations were calculated using the *costDistance* function in the “gdistance” package in R (van Etten 2017).

For measures of current landcover, we used a least-cost transect analysis approach (Van Strien et al. 2012) to evaluate the relationship between landscape resistance and composition, and genetic structure (Cleary et al. 2017; Goldberg and Waits 2010). The least-cost path is based on the conductance surface for the current wetland habitat, with a buffer width of 1 km. For each transect, we calculated the proportion of each landcover type by dividing the number of grid cells that contained a certain landscape type by the total number of grid cells in the transect (function *sample_lsm*, *landscapemetric* package v. 1.5.4, Hesselbarth et al. 2019). Variance inflation factors between explanatory landcover variables were calculated in R (v 4.3.1; R Core Team 2020) using the package “usdm” (Naimi 2015) for the current landcover types proportion of wetland, forest, and developed land.

A total of six candidate MLPE models were run for each species using the “corMLPE” package and function of the same name (v 0.0.3) in R (v 4.0.2). These models included firstly isolation-by-distance (IBD) models which used a cost-surface with equal values (no resistance) between sample sites. Isolation-by-resistance (IBR) models modeled the relationship between genetic distance and least-cost paths based on wetland amount for the periods 1800, 1967, and 2002. Current landscape models tested the relationship between genetic distance and current landscape features (proportion of forest cover, proportion of wetland, and proportion of development) in the intervening landscape between study sites (Table 1). To account for both the

Table 1 Summary of MLPE model hypotheses and description of each. A total of six MLPE models is tested in the snapping, the blanding's, and the spotted turtle

Hypothesis	Description	Dependent Variable	Fixed effects (explanatory variables)	Random effect
Isolation-by-distance: genetic distance best described by geographic distance	Geographic distance calculated as a cost distance across a surface with no resistance between sites	F_{ST}	Cost distance, + Group	Population effects modelled by correlation structure
Isolation-by-Resistance: Genetic distance best described by amount and configuration of wetland habitat in the intervening landscape.	Total cost of least-cost path based on wetland amount through time (c. 1800, 1967, 2002)	F_{ST}	Cost distance + Group	Population effects modelled by correlation structure
Current landscape features: Genetic distance is best described by current landscape features in the intervening landscape	Proportion of landcover type within a 1 km wide transect. Transect length modelled on least-cost path for wetland amount c. 2002	F_{ST}	Forest cover, + Development + Group	Population effects modelled by correlation structure

escarpment as a potential physical barrier and to account for differences in landcover types, all link-based models included a fixed effect assigning the sample location in one of two “groups”: south or north of the Niagara Escarpment (Fig. 1), which were included in all models.

We used the corrected Akaike's Information Criterion (AICc) to identify the best model from the set of candidate models for each species. We fitted MLPE models for the different combinations of landscape features and the models with the lowest AICc (and $\Delta AICc < 2$) were considered the best-fitting models given the genetic structure observed. Sample size in the MLPE models was accounted for by including the number of pairs in each data set. We used the confidence intervals (95%) of estimates from the variables in the model to assess statistical significance. Hence, landscape features with confidence intervals overlapping zero have little evidence of being important to gene flow (Row et al., 2017). R code used for MLPE models was adapted

from codes available in the R “LandGenCourse” package (Wagner 2020).

Results

Genetic diversity

Population-level genetic diversity (Fig. 2) was lowest in the Snapping Turtle, intermediate in the Blanding's Turtle, and highest in the Spotted Turtle (Table 2). The greatest range in values of genetic diversity was found in sites in eastern Ontario among the Blanding's and Snapping Turtle. For the Spotted Turtle, sites with the highest rarefied allelic richness were in southwestern Ontario (sites LH2, GH1, and LE1).

Influence of wetland habitat amount on genetic diversity at the node-level

The amount of wetland habitat at the site was generally poorly correlated with genetic diversity in the Blanding's and Spotted Turtle. In the Snapping Turtle, however, a significant relationship was found between current wetland habitat (c.2002) and measures of rarefied allelic richness and expected heterozygosity (Fig. 3). Ghost landscapes c. 1800 showed a non-significant relationship with rarefied allelic richness ($R^2=0.09$, $F_{1, 6}=1.67$, $p=0.24$), as did wetland loss in 1982 ($R^2=0.35$, $F_{1, 6}=4.72$, $p=0.07$). A significant relationship was found between current wetland habitat (c. 2002) and allelic richness ($R^2=0.55$, $F_{1, 6}=9.54$, $p=0.02$). The strongest correlation among regressions was between expected heterozygosity and current wetland habitat ($R^2=0.73$, $F_{1, 6}=20.2$, $p=0.004$). The same pattern was not found with observed heterozygosity. Table S1 summarize the results for the Snapping Turtle (Supplementary Information).

Influence of latitudinal gradients on genetic diversity at the node-level

Little support was found for the hypothesis that latitude influences genetic diversity for the Snapping, Blanding's or Spotted Turtle (Table 3). Based on linear regressions, we found no relationship between latitude and genetic diversity for the Snapping and Blanding's Turtle, and all measures were a poor fit. A reduction in rarefied allelic richness at higher latitudes was suggested for the Spotted Turtle, but was not significant ($p=0.09$, Table 3).

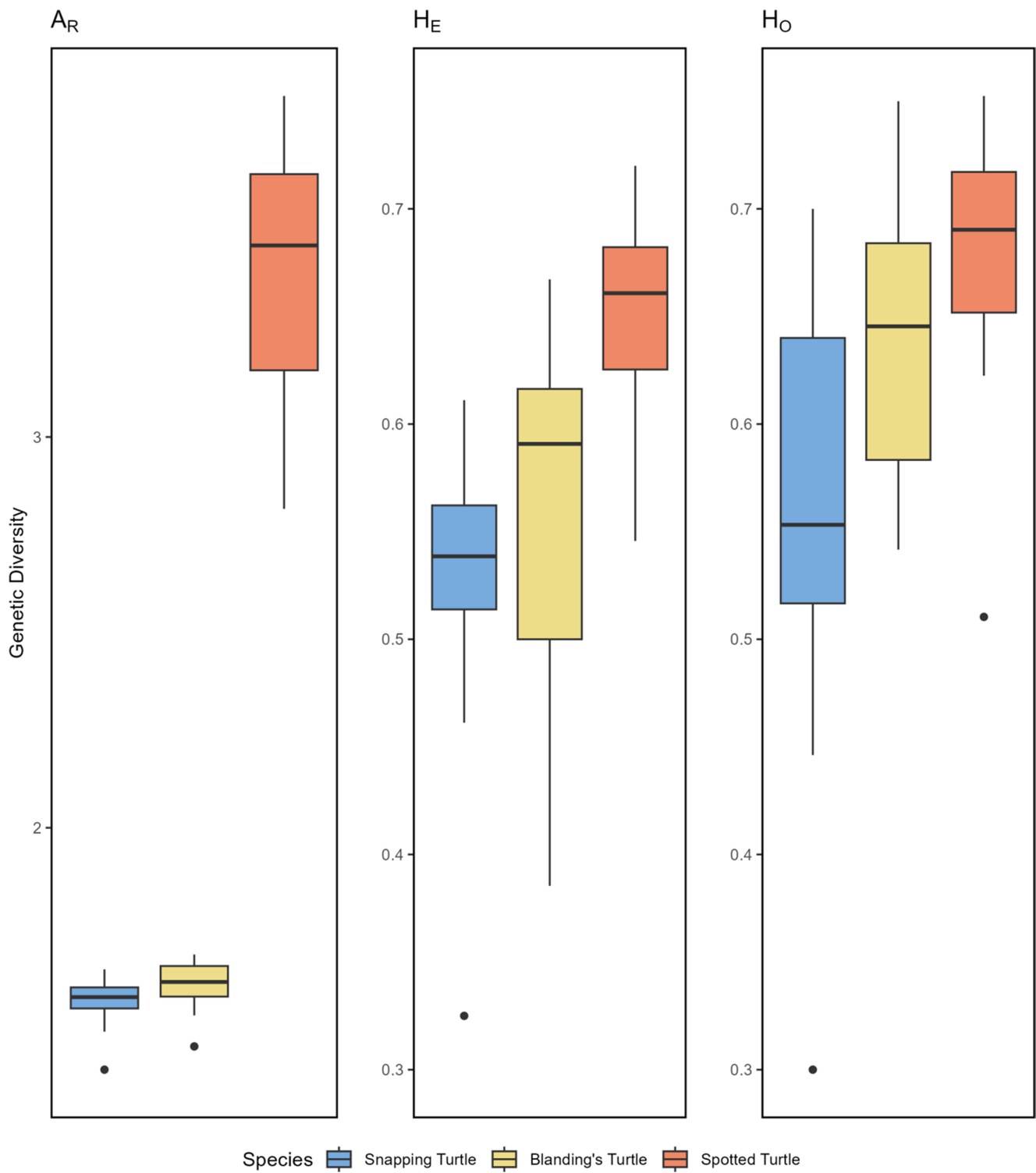


Fig. 2 Box plots of genetic diversity (allelic richness A_R , expected heterozygosity H_E , and observed heterozygosity H_O) in the Snapping Turtle (left bars), Blanding's Turtle (middle bars) and Spotted Turtle (right bars). Boxes indicate the interquartile range (IQR), the horizon-

tal line within each box represents the median genetic diversity value, and whiskers extend 1.5 times the IQR. Dots represent populations with genetic diversity values outside this range

Table 2 Site name, sample size (n), observed (H_O) and expected (H_E) heterozygosity, and rarefied allelic richness (A_R) for samples collected for the Snapping Turtle, the Blanding's Turtle, and the Spotted Turtle in southern Ontario, Canada (2008–2011)

Site	Species	n	H_O	H_E	A_R
ALG	Snapping Turtle	33	0.60	0.56	1.57
	Blanding's Turtle	2	0.58	0.48	1.55
	Spotted Turtle	0	-	-	-
BC	Snapping Turtle	6	0.7	0.61	1.64
	Blanding's Turtle	0	-	-	-
	Spotted Turtle	0	-	-	-
BC1	Snapping Turtle	0	-	-	-
	Blanding's Turtle	4	0.74	0.62	1.66
	Spotted Turtle	0	-	-	-
BP	Snapping Turtle	5	0.7	0.56	1.59
	Blanding's Turtle	0	-	-	-
	Spotted Turtle	31	0.64	0.66	3.46
EL	Snapping Turtle	0	-	-	-
	Blanding's Turtle	2	0.75	0.52	1.59
	Spotted Turtle	0	-	-	-
EO	Snapping Turtle	19	0.64	0.61	1.62
	Blanding's Turtle	20	0.65	0.67	1.68
	Spotted Turtle	0	-	-	-
EO1	Snapping Turtle	0	-	-	-
	Blanding's Turtle	2	0.54	0.39	1.4
	Spotted Turtle	14	0.74	0.63	3.14
EO2	Snapping Turtle	0	-	-	-
	Blanding's Turtle	0	-	-	-
	Spotted Turtle	34	0.75	0.67	3.59
GB1	Snapping Turtle	0	-	-	-
	Blanding's Turtle	0	-	-	-
	Spotted Turtle	6	0.69	0.60	3.18
GB2	Snapping Turtle	5	0.64	0.54	1.57
	Blanding's Turtle	0	-	-	-
	Spotted Turtle	17	0.71	0.65	3.43
GB3	Snapping Turtle	0	-	-	-
	Blanding's Turtle	6	0.67	0.62	1.64
	Spotted Turtle	0	-	-	-
GB4	Snapping Turtle	0	-	-	-
	Blanding's Turtle	9	0.70	0.63	1.65
	Spotted Turtle	0	-	-	-
GH	Snapping Turtle	6	0.47	0.48	1.51
	Blanding's Turtle	1	0.83	0.42	1.56
	Spotted Turtle	0	-	-	-
GH1	Snapping Turtle	0	-	-	-
	Blanding's Turtle	0	-	-	-
	Spotted Turtle	26	0.67	0.71	3.84
GH2	Snapping Turtle	0	-	-	-
	Blanding's Turtle	0	-	-	-
	Spotted Turtle	7	0.51	0.55	2.82
HC	Snapping Turtle	4	0.52	0.54	1.58
	Blanding's Turtle	2	0.58	0.45	1.52
	Spotted Turtle	19	0.62	0.62	3.12
HU	Snapping Turtle	2	0.3	0.33	1.38
	Blanding's Turtle	7	0.65	0.59	1.61
	Spotted Turtle	0	-	-	-
KAW	Snapping Turtle	18	0.56	0.58	1.59
	Blanding's Turtle	0	-	-	-
	Spotted Turtle	0	-	-	-

Table 2 (continued)

Site	Species	<i>n</i>	H_O	H_E	A_R
LE1	Snapping Turtle	27	0.54	0.53	1.54
	Blanding's Turtle	20	0.64	0.60	1.61
	Spotted Turtle	31	0.72	0.72	3.87
LE2	Snapping Turtle	0	-	-	-
	Blanding's Turtle	0	-	-	-
	Spotted Turtle	12	0.69	0.66	3.62
LE3	Snapping Turtle	0	-	-	-
	Blanding's Turtle	14	0.56	0.58	1.59
	Spotted Turtle	0	-	-	-
LH1	Snapping Turtle	28	0.55	0.53	1.54
	Blanding's Turtle	1	0.5	0.25	1.33
	Spotted Turtle	29	0.65	0.66	3.52
LH2	Snapping Turtle	7	0.54	0.46	1.48
	Blanding's Turtle	0	-	-	-
	Spotted Turtle	30	0.72	0.71	3.83
N	Snapping Turtle	7	0.45	0.51	1.54
	Blanding's Turtle	0	-	-	-
	Spotted Turtle	0	-	-	-
mean (SE)	Snapping Turtle		0.554 (0.02)	0.528 (0.02)	1.549 (0.01)
mean (SE)	Blanding's Turtle		0.646 (0.02)	0.523 (0.03)	1.572 (0.02)
mean (SE)	Spotted Turtle		0.677 (0.01)	0.653 (0.01)	3.452 (0.07)

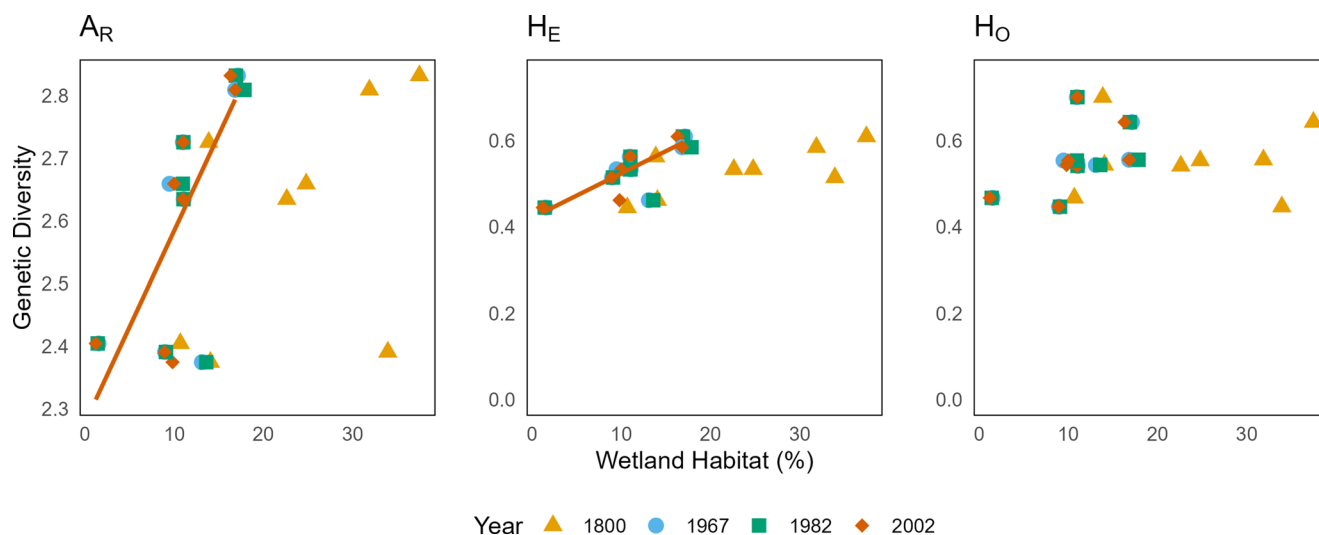


Fig. 3 Univariate regression comparing rarefied allelic richness, expected and observed heterozygosity of the Snapping Turtle to the percentage of wetland habitat amount at the study site circa c. 1800, c. 1967, c. 1982, c. 2002. We identified a significant relationship between

A_R and c. 2002 wetland habitat amount, and H_E and c. 2002 wetland habitat amount. No significant relationships were found between H_O and measures of wetland habitat amount

Isolation-by-distance at the link-level

Genetic distances between pairs of individuals increased with geographic distance in all three species. Evidence of IBD among individuals was found in the Snapping Turtle (Mantel $r=0.06$; $p=0.002$, Fig. 4a), the Blanding's Turtle (Mantel $r=0.34$; $p=0.001$; Fig. 4b), and the Spotted Turtle (Mantel $r=0.14$; $p=0.001$; Fig. 4c).

Influence of landscape on genetic distance at the link-level

Across southern Ontario, the model with the best support of genetic distance between sites in the Snapping Turtle was a current landscape model with the proportion of forest cover (Table 4). The best models of genetic distance in the Blanding's and Spotted Turtle included IBD between sites. The top model for the Spotted Turtle was an IBR model based

Table 3 Results of simple, univariate linear regression models of different measures of genetic diversity as a function of latitude. For each model, the coefficient for latitude, standard error (SE), t -value, p -value, adjusted R^2 , and F -statistic (with degrees of freedom) are reported. No latitudinal trends were identified for any species for any of the diversity metrics at the scale of this study

Species	Model	Coefficient (Latitude)	SE	t	p	Adj R^2	F (df1, df2)
Snapping Turtle	A_R	0.04	0.06	0.62	0.55	−0.07	0.387 (1, 9)
	H_E	0.02	0.02	1.2	0.26	0.04	1.43 (1, 9)
	H_O	0.05	0.03	1.86	0.1	0.2	3.442 (1, 9)
Blanding's Turtle	A_R	0.04	0.03	1.41	0.22	0.14	1.987 (1, 5)
	H_E	0.01	0.01	0.97	0.38	−0.01	0.941 (1, 5)
	H_O	0.03	0.02	1.47	0.20	0.16	2.173 (1, 5)
Spotted Turtle	A_R	−0.27	0.14	−1.94	0.09	0.23	3.753 (1, 8)
	H_E	−0.02	0.01	−1.62	0.14	0.15	2.622 (1, 8)
	H_O	0.01	0.01	0.61	0.56	−0.07	0.377 (1, 8)

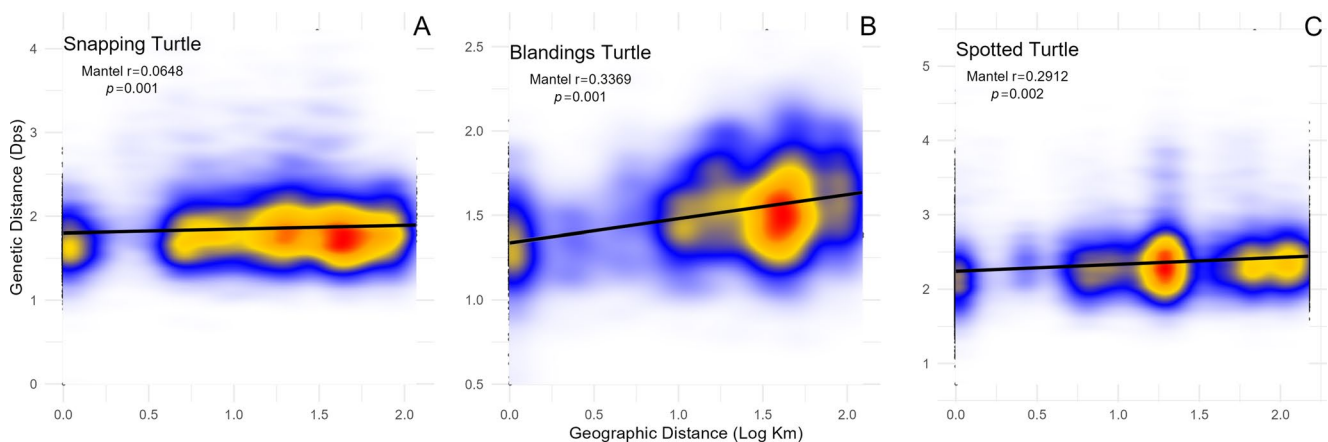


Fig. 4 Genetic distance (proportion of shared alleles, D_{PS}) plotted against geographic distance (log km) for (a) Snapping Turtle (b) Blandings Turtle, and (c) Spotted Turtle. Colours represent the relative density of points, with cooler (blue) representing lower densities. The

black line represents the fitted relationship between distances using linear regression. Mantel coefficient of correlation between geographic and genetic distance, and the associated p -value, is shown

Table 4 Top two MLPE models for the snapping, blanding's and spotted turtle

Species	Model (FST ~)	Δ AICc
Snapping Turtle	Proportion of Forest	0.00
	Proportion of Forest + Proportion of Development	1.07
Blanding's Turtle	Isolation-by-distance	0.00
	Proportion of Development	181.79
Spotted Turtle	Cost distance: Wetland c. 1800	0.00
	Isolation-by-distance	0.652

Table 5 Summary statistics for the best MLPE (maximum likelihood population effect) models selected for each species. In these models, pairwise F_{ST} values were used as a response matrix and different landscape features and predictor matrices. We provide the estimate, the standard error (SE), t -value, and 95% confidence intervals [95% CI]

Model	Estimate	SE	t	95% CI
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Snapping Turtle	−0.00095	0.00039	−1.49	[−0.0017– −0.00019]
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Proportion of Forest				
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Blanding's Turtle	0.01114	0.00263	4.23	[0.00598– 0.01630]
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Isolation-by-distance				
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Spotted Turtle	0.00996	0.00232	4.29	[0.00541– 0.01451]
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Cost Distance Wetland				
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c. 1800				
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on wetland extent c. 1800 and was within a $\Delta AICc$ of 2 with model of IBD. Following Row et al. (2017), we use confidence intervals from the variables in the models to identify those contributing most to landscape resistance. Predictors with confidence intervals that overlap 0 are considered to be of little importance to gene flow. Confidence intervals for the top model in the Snapping Turtle did not overlap zero, indicating that proportion of forest cover has some evidence of influencing gene flow (Table 5). As the values for the Snapping Turtle are negative, forest cover is indicative of more gene flow. Confidence intervals also did not overlap zero in the case of the Blanding's Turtle top model, IBD, and the Spotted Turtle top model, based on historic (c. 1800) wetland habitat extent (Table 5). Geographic distance impedes gene flow for both the Blanding's and the Spotted Turtle. The configuration of wetland habitat c. 1800 also contributed to genetic differentiation in the Spotted Turtle.

Discussion

We investigated the relative contribution of historic and current drivers of genetic diversity and structure in three sympatric long-lived turtle species. Understanding how landscapes influence genetic diversity among sympatric species is important for landscape-scale conservation but is difficult to uncover in long-lived species.

Patterns of genetic diversity differ among species

Measures of genetic diversity were comparable among species, despite differences in abundance. The species with the lowest measures of genetic diversity was the Snapping Turtle. A phylogeographic sampling of the Snapping Turtle in populations in the south-eastern United States also noted lower than expected genetic variation (Walker et al. 1998).

A negative correlation is expected between genetic diversity and latitude in post-glacial landscapes due to founder effects during northward range expansion (Galbraith 2008). We therefore expected to identify latitudinal trends in genetic diversity. Given the climatic constraints experienced in northern populations, it has been predicted that habitats such as overwintering sites may be limited and these constraints make it difficult for turtles to compensate for population declines (Brooks et al. 1991; Brown and Brooks 1994). At the population-level, the effects of these processes were not detectable in this study. In the Spotted Turtle, we did detect a slight trend showing lower levels of genetic diversity at higher latitudes, but this trend was not statistically significant. A comparison of this Spotted Turtle dataset (Davy and Murphy 2014) with measures of genetic diversity in the southern half of Spotted Turtles range in the southeastern United States revealed comparable levels of genetic diversity and therefore no apparent effect of latitude at a wider scale on genetic diversity (Brown et al. 2024). Most models comparing genetic diversity to latitude were a poor fit (Table 3).

Isolation-by-distance

The strongest IBD trend was seen in the relatively vagile Blanding's Turtle (Fig. 4). Previous studies that have examined IBD in Blanding's Turtle have used population-based measures (Davy et al. 2014; Jordan et al. 2019; Mockford et al. 2007; Reid et al. 2017) with varying conclusions regarding the effects of distance on population genetic structure. Comparing the proportion of shared alleles among individuals, McCluskey et al. (2022) found landscape resistance to be a stronger driver of genetic differentiation than IBD on a smaller spatial scale than our study. The geographic range of the Blanding's Turtle extends across North America through

the Great Lakes region, with disjunct populations in New England, New York, and Nova Scotia. We also identified an IBD trend in the Spotted Turtle, which we expected due to their relatively lower vagility. Fine-scale spatial genetic structure has previously been observed in the Spotted Turtle, with individuals at distances < 25 km apart showing higher spatial genetic structure (Liebgold et al. 2023).

In this study, and others (Frankham 2010, 2015) genetic similarity among individuals in close proximity is an important consideration for a species of conservation concern whose populations are often separated by greater distances. Given contrasting findings among studies addressing landscape drivers of genetic differentiation in the Blanding's Turtle, and patterns of isolation by distance in the Spotted Turtle, regional management of this species and other endangered species is warranted. Despite the relatively higher abundance, the Snapping Turtle in southern Ontario exhibits IBD, highlighting the dispersal limitation of a more abundant species inhabiting a fragmented landscape.

Isolation of subpopulations with increasing land cover change can drive increased inbreeding and put remnant subpopulations at a greater risk of inbreeding depression (Duntsch et al. 2023). Despite the wider distribution and apparently greater connectivity of Snapping Turtle subpopulations in our study area, compared to the other two species, our data showed that even this species exhibits limited gene flow between relatively proximate sampling areas. Our results and those of related studies can inform the management of these three species at multiple spatial scales, ensuring range-wide persistence and conserving adaptive potential within smaller management units.

Habitat loss and declines in genetic diversity

In the Snapping Turtle, A_R and H_E were the highest at sites with a higher percentage of remnant wetland habitat. This significant relationship was unexpected, given the generation time is longest in the Snapping Turtle and the genetic effects of habitat loss take time to manifest (Schlaepfer et al. 2018). This result contrasted with the more specialist Blanding's and Spotted Turtle species, in which we found no correlation between measures of genetic diversity and the amount of wetland habitat at sample sites. The Snapping Turtle is considered widespread and abundant across Ontario but a decline in population numbers has occurred (COSEWIC 2008). One possible explanation for lower genetic diversity at sites that have lost more wetland habitat is population declines at these more degraded sites. Relatively more degraded sites are found in southwestern Ontario where more wetlands have been lost relative to sites north of the Niagara escarpment. As habitat is lost, genetic

diversity is expected to decrease as declining habitat supports smaller populations.

Thus, our results could also indicate a population bottleneck in southern Ontario Snapping Turtles, although we recognize that further research would be required to confirm or refute this. In a population bottleneck, allelic richness decreases first as some alleles are randomly lost from a population, followed by a lagged decrease in heterozygosity as the gene pool decreases (Nei et al. 1975). A recent simulation study demonstrates a trade-off between A_R and H_O in fragmented habitats (Hill et al. 2023) similar to that experienced by Snapping Turtles in southern Ontario. Our results (lower genetic diversity in the southern most sites) tentatively support the hypothesis that Snapping Turtle populations have experienced greater declines in areas with most habitat loss and degradation.

Influence of landscape on genetic distance

We found evidence that dispersal limitations and anthropogenic land cover change restricts gene flow among turtle populations. Both Blanding's and Spotted Turtles, two specialist species, were influenced by geographic distance and historic configurations of wetland habitat, respectively, while a model of a current landscape feature was supported in the generalist species, the Snapping Turtle (Table 5). Previous clustering analysis on the Spotted Turtle (Davy and Murphy 2014) showed the isolation of some neighbouring sites while other sampling sites similar distances apart continue to form single genetic clusters. This result suggests that these groups were once part of a larger and more connected population spread across the intervening landscape. The top model for specialist species indicates that cost of movement based on historic (c. 1800) wetland amount and isolation by distance limited gene flow in both species. An analysis of the spatial and temporal patterns of wetland loss in the region found that wetland habitats near Lake Ontario and Lake Erie were likely already impacted by European settlers by 1800 (Penfound and Vaz 2022). This is only about 10 generations ago for the Spotted and Blanding's Turtles, which have a similar generation time of >25 years. Though, simulation studies suggest this may be enough time to detect barriers to gene flow (Blair et al. 2012), this signal may be masked by other factors that contribute to genetic structure. A higher proportion of forest cover between sample sites indicated more gene flow in the Snapping Turtle. Given genetic distance is lower in areas with higher forest cover in the intervening landscape, increasing genetic connectivity and diversity for the Snapping Turtle is best supported by increasing both forest and wetland habitat. This is particularly important in southwestern parts of the study

area where significant historic wetland and forest habitat has been lost.

Conservation implications

Turtles are among the most vulnerable taxa to anthropogenic change (Böhm et al. 2013; Lovich et al. 2018) and previous studies have highlighted the negative impacts of landscape change on genetic diversity patterns (Reid et al. 2017; Turcotte et al. 2022). Wetland and forested habitats continue to change in southern Ontario, Canada, and further habitat loss will degrade the future genetic diversity of wild populations. Given the continued decline in genetic diversity among species inhabiting landscapes where human influence is greatest, understanding how landscape change has influenced the genetic composition of long-lived species like turtles is of concern for future conservation management.

Overall, our study shows that these sympatric species are isolated, to differing degrees, by geographic distance. These long-lived species have limited dispersal and inhabit increasingly fragmented habitats which poses a threat to their long-term persistence. Our results further support the need to assess generalist species, in addition to focusing on endangered species. Though the Snapping Turtle has less specific habitat requirements, they are negatively impacted by habitat loss and subsequent declines in genetic diversity. Further loss or degradation of habitat across southern Ontario will further limit gene flow among populations, but it will likely be many generations before these effects are apparent in measures of genetic diversity in sympatric species. This should also be considered when measuring conservation success, as a temporal lag in these species' responses to conservation action is also probable (Watts et al. 2020). Lack of genetic variability is not likely a threat to these long-lived species and conservation efforts should therefore focus on demographic threats. Yet, our results highlighting a potential bottleneck in the Snapping Turtle, the most abundant and long-lived, demonstrates that these species are still at risk of declines in genetic diversity. Increasing population size can prevent further losses of genetic diversity in threatened populations (Frankham et al., 2002) and restoration and protection of current habitat remains critically important for long-term conservation.

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Author contributions All authors contributed to the study conception and design. Data collection was performed by C. M. D. and data analysis was performed by N. M. W. The first draft of the manuscript was written by N. M. W. and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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